



THE UNITED REPUBLIC OF TANZANIA
THE PRIME MINISTERS
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MBEYA CITY COUNCIL
MBEYA CITY JOINT EXAMINATION
FORM THREE BIOLOGY
CHEMISTRY 1



032/01

TIME 3:00 HRS

MARCH, 2026

Instructions

1. This paper consists of section A, B and C with total of eleven (11) questions.
2. Answer all questions in section A and B and two questions from section C
3. Section A carries 16 marks section B carries 54 marks and section C carries 30 marks
4. All writing should be in blue or black ink except for diagrams that must be drawn in pencil
5. Remember to write your candidate examination number at the top of every page.
6. Cellular phones and any unauthorized materials are not allowed in the examination room.
7. The following constant may be used.
 - GMV at S.T. $P=22.4 \text{ dm}^3$
 - Atomic Mass= H=1, C= 12, O= 16, Na=23, S=32, Cl= 35.5, K=39, Fe=55.8, Ca= 40, Mg =24, Ag = 108, Cu = 63.5,
 - 1Faraday = 96500 Coulombs
 - 1Litre= $1\text{dm}^3 = 1000\text{cm}^3$
 - Avogadro's constant, (A)= 6.02×10^{23}

GEPAM science hub

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SECTION A (16 Marks)

Answer **all** questions in this section.

1. For each of the items (i) – (x), choose the correct answer from the given alternatives and write its letter in the table provided below.

- (i) During volumetric analysis in the school chemistry laboratory the solution added to the burette by a first student is referred to as:
- A. Acid and base
 - B. Secondary solution
 - C. Analyte
 - D. Titrant
 - E. Alkaline solution.
- (ii) The valency of an element found at Chunya mining industry with atomic number 10 is :-
- A. 2
 - B. 1
 - C. 3
 - D. 0
 - E. 4
- (iii) Which method could be used to separate the products in the following chemical reaction?
- $$\text{Pb(NO)}_2(\text{aq}) + 2\text{KI}(\text{aq}) \rightarrow \text{PbI}_2(\text{s}) + 2\text{KNO}_3(\text{aq})$$
- A. Condensation
 - B. Filtration
 - C. Chromatography
 - D. Distillation
 - E. Separating funnel
- (iv) 10 cm³ of 0.4M sodium hydroxide are added to 40cm³ of 0.2M hydrochloric acid. The resulting mixture will be:-
- A. Alkaline
 - B. Neutral
 - C. Acidic
 - D. Strong alkaline
 - E. Weak acidic
- (v) Mr. O. mixed two solutions in a boiling tube at 20°C. After mixing the two solutions the new products were found to have a temperature of 10°C. This shows that the reaction was;
- A. An exothermic
 - B. An endothermic
 - C. Enthalpy
 - D. neutralization
 - E. combustion

- (vi) What is the IUPAC name of H_2SO_3
- A. Sulphuric (IV) acid
 - B. Sulphuric (VI) acid
 - C. Sulphuric acid
 - D. Hydrogen sulphate trisulphate
 - E. Hydrogen
- (vii) Which of the following sets of process use a gas that ignites with a pop sound when a lighted splint is passed through it?
- A. Ballon filling, dividing rocket, diving and welding
 - B. Hardening of oil, ballon filling
 - C. Hardening of oil, ballon filling and and diving
 - D. Fueling of welding and diving
 - E. Fueling car, diving and welding.
- (viii) Mr. Machepe Abdallah was demonstrating an experiment by dissolving sugar in water until the solute was not dissolving any more, the resulting mixture is;
- A. Emulsion
 - B. Saturated
 - C. Suspension
 - D. Unsaturated
 - E. Super Saturated solution
- (ix) Image your brother was preparing dinner in the kitchen, he started chopping potatoes into small pieces to make chips, what was the purpose of this action?
- A. Make them delicious
 - B. Make them dry
 - C. Shorten the cooking time
 - D. To increase the amount of potato chips
 - E. To increase concentration of potatoes.
- (x) Which of the following substances where its dissolution in water causes permanent hardness of water?
- A. Calcium sulphate
 - B. Magnesium carbonate
 - C. Sodium Chloride
 - D. Calcium Carbonate
 - E. Sodium Carbonate

2. Match the type of reaction in **List A** with a correct response in **List B** by writing the letter of the correct response below the corresponding item number in answer sheet provided .

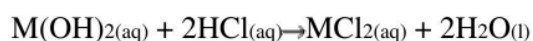
LIST A	LIST B
(i) Decomposition reaction	A: Shows only chemical species that involved in the reaction.
(ii) Displacement reaction	B: $F_2 + 2HCl \rightarrow 2HF + Cl_2$
(iii) Net ionic equation	C: Complex chemical compound broken into simple molecules.
(iv) Neutralization reaction.	D: Combination reaction.
(v) Complete ionic equation	E: Double decomposition reaction.
(vi) $2H_2(g) + O_2(g) \rightarrow 2H_2O$	F: Both spectator ions and reacting species are included.
	G: Found both sides of the chemical equation.
	H: Molecular formula
	I: Only hydroxide ions and hydrogen ions are involved

SECTION B (54 Marks)

Answer **all**

3. (a) Hydrogen is most useful element in the world, you as chemist explain how hydrogen is made in the laboratory and industry (use chemical equations)
- (b) Explain how does hydrogen react with copper (II) oxide? (Chemical equation is necessary)
- (c) Hydrogen is a promising fuel for the future. Justify this statement by giving two (2) reasons.
- (d) Most uses of hydrogen are dictated by its properties. Explain this statement by giving two reasons

4. The hydroxide of metal M react with HCl as follows;-



It was found out that 5.8g of $M(OH)_2$ react with 0.2 M HCl. Calculate the

- (i) Atomic mass of metal M
- (ii) Suggest the name of metal M.
5. Anna, Nandii, Godfrey, miza, Magreth and Ester Performed an experiment to determine hardness of water by using soap and the following water sample.

Water sample A = water from longa , Water sample B = water from itizi, Water sample C = water from kimo and Water sample G = water from uloza .

Sample	VOLUME OF SOAP SOLUTION USED (cm ³)		
	Untreated water	boiled water	Water passed in the ion exchanger
A	12	1.8	1.8

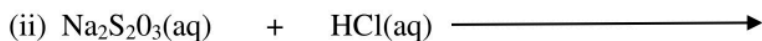
B	17	17	1.7
C	26	20	1.8
D	1.6	1.6	1.6

- (a) Which water sample is hardest? Give reason for your answer.
 - (b) Which water sample behave as distilled water? Explain.
 - (c) Name a chemical substance that could be that cause of hardness in
 - (i) water sample A
 - (ii) water sample B
 - (d) Which problems may face people who are using water sample of A, B and D?
6. (a) Form three students visited chemistry laboratory and laboratory technician insisted them to obey all statements that explain what to do or not to do before, during and after practical activities. Give three (3) reasons why should these statements be obeyed by students in the laboratory?
- (b) Give one reason to support each of the following statements
- i. A laboratory should be equipped with working fire extinguisher's
 - ii. A laboratory should have large windows
 - iii. A laboratory should have fume chamber
- (c) i. In which other areas do you find the safety signs? (two (2) points)
- ii. Briefly explain why are the safety signs so important in the laboratory?
7. (a) What volume of hydrogen gas is produced at STP when 0.65g of zinc reacts completely with excess hydrochloric acid?
- (b) What mass of sulphuric acid is found in 400 cm³ of its 0.1M aqueous solution?
- (c) What is the molarity of :-
- (i) solution containing 5.3g of sodium carbonate in 100ml of the solution?
 - (ii) solution containing 80g of sodium hydroxide in 2 litre?
8. One of the form one student decided to separate the mixture of water and sand which are mixed accidentally, use this information to respond on the questions below;
- i. What are the apparati/equipment's which can be used to separate the above mixture (Give three points).
 - ii. By using a well labeled diagram explain how those two components can be separated by this student.
 - iii. Briefly explain how are solutions, suspensions and emulsions important in our life? (Give three points)
 - iv. Form three students observed that water is changing to vapour after being boiled to 100°C. You as chemist, explain how the process of changing water to vapour is used at home. (Give three (3) points).

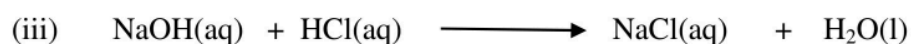
SECTION C (30 Marks)

Answer **two (2)** questions in the section.

9. (a) Complete and Balance the following reaction and write the ionic equation.



(b) State the types of reaction in the following compound



10. (a) One of form three students was assigned a task of cutting the branches of trees around school laboratory. Accidentally a bee sting a student on his arm. Perhaps a teacher assisted him by applying soap on it.

- i. Mention the type of reaction which was expected to take place when soap were applied
ii. Give out other four (4) application of the reaction occurred in a (i) above in our daily life

(b) With the aid of a balanced chemical equation, suggest the best methods for preparing the following salts in the laboratory.

- (i) Lead (II) sulphate
- (ii) Copper (II) nitrate
- (iii) Calcium Chloride
- (iv) Calcium Carbonate
- (v) Barium Sulphate

11. Most of farmers in the KATONDO Village were interested to know how the knowledge of chemistry is necessary in agricultural activities. Assume you are chemist; educate these farmers on the application of chemistry in agriculture by giving five points

PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENTS
MBEYA CITY COUNCIL

FORM THREE MID – TERM JOINT EXAMINATION

032/01

CHEMISTRY 1 MARING SCHEME

SECTION A (16 Marks)

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1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
D	D	B	C	B	A	B	E	C	A

2.

i	ii	iii	iv	v	vi
C	B	A	I	F	D

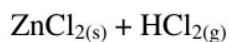
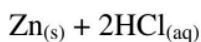
1mark@ = 06 marks

3.

In laboratory

Reaction of dilute hydrochloric acid with Zinc metal

01½ mark

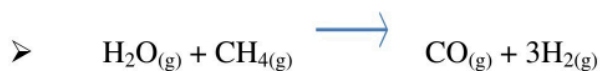


½ mark

In industry



By steam reforming , Steam reacts with methane to produce carbon monoxide and hydrogen gas



Copper (ii) oxide is reduced to copper metal as shown below:-



c) i. Hydrogen does not cause environmental pollution

ii. Hydrogen can be produced from renewable resources eg. Water

d)

Property	Uses
i) Reducing agent	Manufacturing margarine
ii) Lighter than air	Filling weather balloons
iii) Highly flammable	Preparation of water gas
iv) Readily react with other substances	Manufacturing of ammonia

4.a) i. Data given

Mass of $M(OH)_2$, $m = 5.8g$

Molarity of HCl , $M = 0.2M_T$.

- Required, atomic mass of metal M , M_r .
- Consider the following equation



1mol of $M(OH)_2 \equiv 2\text{mol of } HCl$

$X? \equiv 0.2\text{mol of } HCl$

$$X = \frac{0.2\text{mol}}{2} = 0.1\text{mol}$$

Moles of $M(OH)_2 = 0.1\text{mol}$

From

Moles = $\frac{\text{mass}}{\text{Molar mass}}$

Molar mass

$$0.1\text{mol} = \frac{5.8g}{M_r}$$

Molar mass = $58g/mol$

- In magnitude molar mass is equal to relative atomic mass

Hence

Atomic mass of $M(OH)_2 = 58$

$M(OH)_2 = 58$

$M + 32 + 2 = 58$

$M = 58 - 34 = 24$
= 24

Atomic mass of $M(OH)_2 = 24$

ii. The name of metal M is magnesium

4 (b) Data given

1M

Volume of acid, $V_a = 25\text{cm}^3$

Molarity of acid, $M_a = 1\text{M}$

Molarity of base, M_b

=

01 mark

Consider the following equation



From

$$\frac{M_a V_a}{M_b V_b} = \frac{n_b}{n_a}$$

01 mark

$$\frac{1 \times 25}{1 \times V_b} = \frac{1}{2}$$

$$\frac{25}{V_b} = \frac{1}{2}$$

$$V_b = 12.5\text{cm}^3$$

☐ The volume capacity of the pipette used = 12.5cm^3 **01 mark**

c) The burette should be cleaned and rinsed in order to avoid contaminations that may interfere with the reaction. **2 marks**

5. a) water sample C is hardest, because it remains hard even after boiling.

2marks

b) water sample D behave as distilled water because it use Little volume of soap solution. **2 marks**

It means that water contains no soluble substances.

c) (i) Calcium hydrogen carbonate or magnesium hydrogen carbonate. **1marks**

(ii) Calcium sulphate or magnesium sulphate **2 marks** d)

- using more soap during washing

- destruction of boilers, hot water pipes and kettles.

Any 3 points, 1 mark @ = 2marks

6. a) Laboratory rules should be obeyed by everyone because laboratory rules help;

- To have successful practical activities

- To ensure safety

- To protect equipments/apparatii in the laboratory

- To avoid contaminations **(Any 3points,1mark@=3marks)**

b) i. A laboratory should be equipped with fire extinguisher so as to extinguish harmful fire in case there is fire accident.

ii. A laboratory should have large windows for proper and good ventilation

iii. A laboratory should have fume chamber for preparing poisonous gases

(1mark@=3marks)

c) i. - In hospital

- In petrol filling station

- In chemical industries

(Any 3points, 00¹mark@=1 ¹ marks)

ii. This is because safety signs ensure safety in the laboratory during practical activities. Hence safety signs warn laboratory users on the hazards of chemical and advice laboratory users to handle chemicals carefully

$\frac{1}{12}$ marks)

7. a) given data.

Mass = 0.65g

Molar mass = 98g/mol.

Volume = needed **$\frac{1}{2}$ marks**

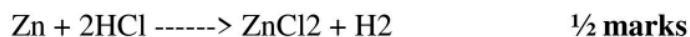
Firstly, calculate number of moles then convert into volume

$$n = \frac{\text{mass}}{\text{Molar mass}}$$

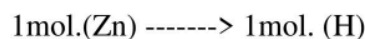
$$n = \frac{0.65}{65.5 \text{ g/mol}}$$

$$= 0.0099 \text{ mol.}$$

From the eqn.



This means



$$? = 0.0099 \times 1$$

1

$$= 0.0099 \text{ mol(H)} \quad \frac{1}{2} \text{ marks}$$

Convert the obtained moles of hydrogen into volume.

From.

$$n = \frac{\text{volume}}{\text{Molar gas volume}}$$

gas volume

$$\text{Volume} = n \times \text{molar gas volume} \quad \frac{1}{2} \text{ marks}$$

$$= 0.0099 \times 22.4$$

$$= 0.222 \text{ dm}^3. \quad \frac{1}{2} \text{ marks}$$

(b) given data.

$$\text{Molarity} = 0.1 \text{ M}$$

$$\text{Volume} = 400 \text{ cm}^3$$

Mass = needed

$$\text{Molar mass} = 98 \quad \frac{1}{2} \text{ marks}$$

Firstly find the number of moles in 400 cm^3 then convert moles into mass.

Remember that Molarity means the number of moles of solute that dissolved to make one litre of the solution.

$$0.1 \text{ mol.} \rightarrow 1000 \text{ cm}^3$$

$$? \leftarrow 400 \text{ cm}^3$$

$$? = 0.1 \times 400$$

$$1000$$

$$0.04 \text{ mol (in } 400 \text{ cm}^3) \quad \frac{1}{2} \text{ marks}$$

Convert 0.04 mol. Into mass.

mass

$$n = \frac{\text{mass}}{\text{Molar mass}}$$

Molar mass

$$\text{Mass} = n \times \text{molar mass}$$

$$= 0.04 \times 98$$

$$= 3.92\text{g.} \quad \frac{1}{2} \text{ marks}$$

(c) (i) data given.

$$\text{Volume} = 100\text{cm}^3$$

$$\text{Mass} = 5.3\text{g}$$

Molarity = required

$$\text{Molar mass} = 106\text{g/mol.} \quad \frac{1}{2} \text{ marks}$$

But molarity = number of mole in one litre of the solution. Also
concentration(g/dm³)

$$\text{molarity} = \frac{\text{concentration(g/dm}^3\text{)}}{\text{Molar mass g/mol}} \quad \frac{1}{2} \text{ marks}$$

Convert 5.3g in 100cm³ into g/L(litre)

$$5.3\text{g} \text{ -----} > 100\text{cm}^3 ?$$

$$<\text{-----} 1000\text{cm}^3$$

$$? = 5.3 \times 1000$$

$$100$$

$$= 53\text{g/dm}^3 \quad \frac{1}{2} \text{ marks}$$

concentration(g/dm³)

$$\text{molarity} = \frac{\text{concentration(g/dm}^3\text{)}}{\text{Molar mass g/mol}}$$

$$\text{molarity} = \frac{53(\text{g/dm}^3)}{106(\frac{\text{g}}{\text{mol}})}$$

$$= 0.5\text{M} \quad \frac{1}{2} \text{ marks}$$

(ii)

data given.

$$\text{Volume} = 2\text{L} (2000\text{cm}^3)$$

$$\text{Mass} = 80\text{g}$$

Molarity = required

$$\text{Molar mass} = 40\text{g/mol.} \quad \frac{1}{2} \text{ marks}$$

But molarity = number of moles in one litre of the solution. Also
=

$$\text{molarity} = \frac{\text{concentration(g/dm}^3\text{)}}{\text{Molar mass g/mol}}$$

s Convert 80g in 2000cm³ into g/L(litre) or dm³

$$80\text{g} \text{ -----} > 2000\text{cm}^3$$

$$? <\text{-----} 1000\text{cm}^3$$

$$? = 80 \times 1000$$

2000

$$= 40 \text{ g/dm}^3 \quad \frac{1}{2} \text{ marks}$$

26

Molarity = concentration

Molar mass

$$= 40$$

40

$$= 1 \text{ M} \quad \frac{1}{2} \text{ marks}$$

Total 8 marks

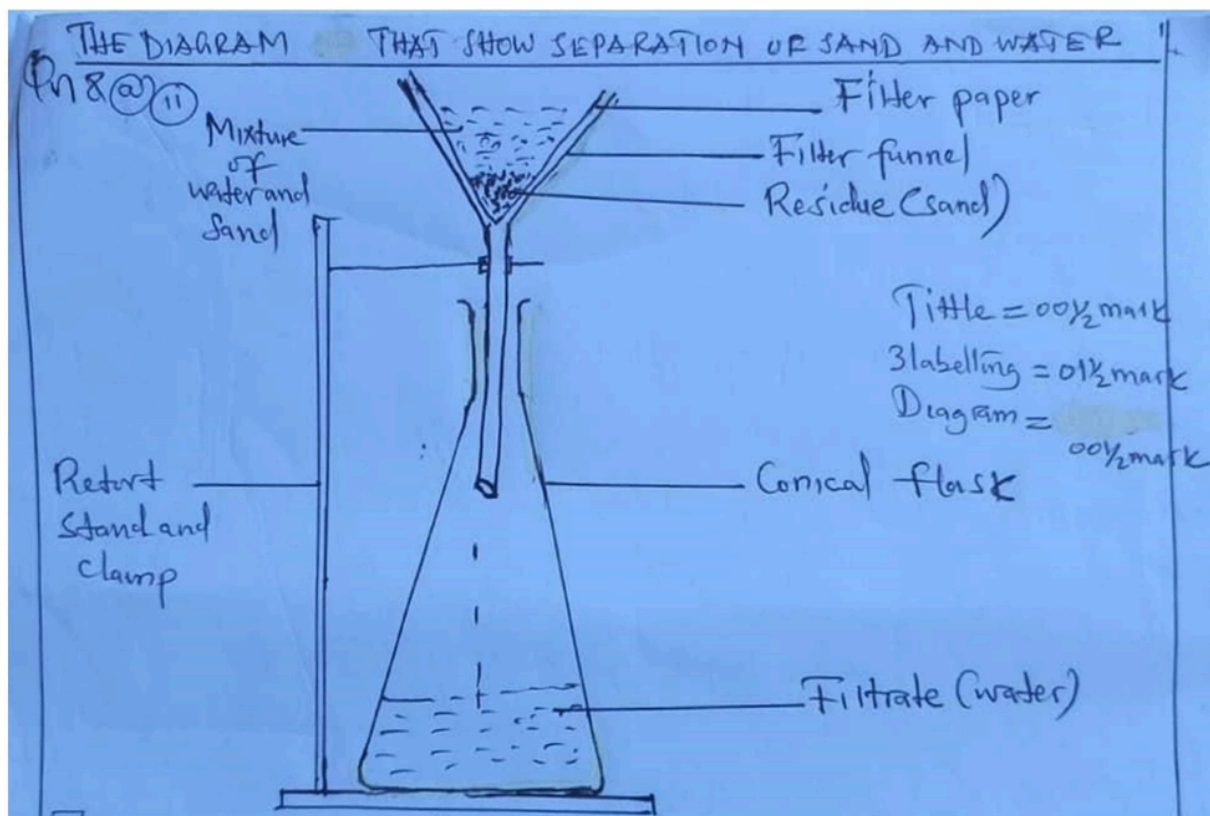
8a) i. Conical flask/Flat = bottomed flask

- Filter funnel

- Retort stand and clamp

- Filter paper (Any 3 points, 1 mark @ = 1 1 marks)

2 2



The sand is separated from the liquid using a porous filter such as filter paper. The sand will remain in the filter paper as residue and water will pass through filter into the conical flask.

8b) - Solution → used in cleaning eg. Soap solution

- Suspensions → Used as drug for treatment of diseases

- Emulsion → Used in decoration eg. Paint
 $\frac{1}{2}$ mark @ = 1 mark

c) - Used in drying crops

- Used in drying clothes
- Used in drying mopped areas
- Used in drying washed hairs

Any 3 points, 1 mark @ = 3 marks

SECTION C: (30 Marks)

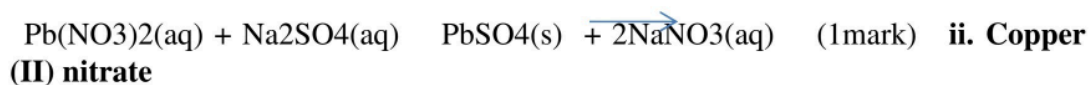
9.

10a) i. Neutralization reaction (1 mark) ii. – Used in relieving indigestion

- Used in controlling soil PH
- Used in treating factory wastes (Any 4 points, 1 mark @ = 4 marks) - Preventing formation of acid rain - Manufacturing of fertilizers

b) i. Lead (II) sulphate

It is prepared by precipitation reaction (1 mark)

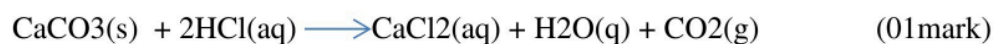


It is prepared by reacting dilute HNO_3 with metal carbonate
 (1 mark)



iii. Calcium Chloride

It is prepared by reacting dilute HCl with metal carbonate (01 mark)



iv. Calcium Carbonate

It is prepared by precipitation reaction (01mark)



v. Barium sulphate

It is prepared by precipitation reaction (01mark)



11. i. Introduction

- Meaning of chemistry $\frac{1}{2}$ mark (1 $\frac{1}{2}$ mark)
- Meaning of Agriculture $\frac{1}{2}$ mark (1 $\frac{1}{2}$ mark)

ii. Main body

- It helps to make fertilizers
- It helps to produce pesticides (**Any 5points,2mark@=10marks**)
- It helps to produce animal vaccine
- It helps to produce weed killers
- It helps to make insecticides

iii. Conclusion (2mark)